Cicy K. Agnes

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Education

M.Sc. Computer Science — May 2023 – Present
Universität des Saarlandes — Saarbrücken, Germany

B.Tech. Computer Science and Engineering — Aug 2018 – Sept 2022
Cochin University of Science and Technology — Kochi, India

Research Experience

Research Assistant: Trustworthy NLP — 05/2026 – Present
German Research Center for Artificial Intelligence (DFKI), Language Technology Lab — Berlin (remote)

- Evaluate multilingual language models for robustness and cultural bias; design and run debiasing experiments for underrepresented languages within the TRAILS project.
- Build reproducible Python evaluation pipelines for benchmarking.

Junior Researcher: AI Systems & Decision Support — 06/2024 – 03/2026
German Research Center for Artificial Intelligence (DFKI), Smart Service Engineering Lab — Saarbrücken, Germany

- Developed energy-efficient AI methods for sustainable data (ESCADE Project). Papers: [3, 4, 7].
- Concept and design of Artificial Mental Models (FedWell Project). Papers: [5, 6, 10, 12]
- **Developer lead, EAVE:** designed and implemented an AI-based information system for operational decision support, data ingestion pipelines, predictive modelling, and an interactive dashboard letting operators explore scenarios and act on forecasts of energy, CO_2 and cost.
- Co-built an autonomous agent for dynamic resource management acting on live observable signals only; evaluated in simulation on production-scale Azure inference traces, cutting GPU energy use by up to **41.2%** within latency constraints.
- Ran research studies independently end to end, problem framing, architecture, implementation, evaluation, delivery, and mentored junior team members.
- Presented research at international conferences and to industry stakeholders.
- Supported research proposal writing: technical narratives, work-package planning.

Research Assistant: Knowledge Graphs & Applied AI — 04/2023 – 05/2024
German Research Center for Artificial Intelligence (DFKI), Smart Service Engineering Lab — Saarbrücken, Germany

- Papers: [8,9, 11, 13]
- Developed an end-to-end method extracting knowledge graphs from full-text plan descriptions and generating new plans from user-supplied requirements and context, graph extraction, representation and planning in one pipeline.
- Developed a method, combining semantic priming with structured knowledge graphs to detect and mitigate factual errors in language-model outputs.
- Data processing, model training, benchmarking and structured error analysis; supported national funding proposals.

Talks & Demonstrations

- Live demonstrator for DFKI, **Hannover Messe**: energy-aware AI operations for data centres. 2025
- Live demonstrator for DFKI, **Berlin GreenTech Festival**. 2025
- Collaborative Meetings: DZNE Bonn, Deutsche Telekom Munich, LIST Luxembourg etc..
- Conference presentations: **NeurIPS, AAAI Symposium, Dagstuhl**. 2024–2026

Publications Google Scholar

[1] **Jury of Unequals: Improving Ranking in LLM-as-a-Judge Frameworks**
C. K. Agnes, D. A. Selby, S. Vollmer, G. Großmann
NeurIPS, JUDGe Workshop (Submitted for review), 2026.

[2] **Contrastive Representational Decoding for Hallucination Detection**

C. K. Agnes
NeurIPS, Women in Machine Learning Workshop (Submitted for review), 2026.

[3] **Safety-Constrained Contextual Bandit for Dynamic Power Management**
Y. Esfandiari, K. Rajshekar, **C. K. Agnes**, H. Stein, S. Janzen, W. Maass
IJCAI, SuRe Workshop, 2026.

[4] **Towards Decision Support Systems for Cost-effective and Energy-efficient AI Operations in Data Centers**
H. Stein, S. Janzen, **C. K. Agnes**, D. T. Ninh, W. Maass
International Conference on Advanced Information Systems Engineering (CAiSE), Springer, pp. 394–413, 2026.

[5] **AI in Rehabilitation — Application of Artificial Mental Models for Personalized Medicine**
S. Janzen, P. Saxena, **C. K. Agnes**, W. Maaß
Bundesgesundheitsblatt — Gesundheitsforschung, Gesundheitsschutz 68(8), pp. 889–897 (also published in German), 2025.

[6] **FEDWELL: Life-Long Federated User and Mental Modeling for Health and Well-being**
S. Janzen, P. Saxena, **C. K. Agnes**, M. E. U. Khan, A. Gomaa, M. Feld, A. Zenner et al.
CAiSE Research Projects Exhibition, 2025.

[7] **ESCADE: Energy-efficient Artificial Intelligence for Cost-effective and Sustainable Data Centers**
S. Janzen, H. Stein, K. Trinley, **C. K. Agnes**, V. Jain, K. Rajeshkar, N. Shenoy et al.
CAiSE Research Projects Exhibition, 2025.

[8] **Semantic Priming via Knowledge Graphs to Analyze and Treat Language Models' Honest Lies**
C. K. Agnes, M. R. Rahman, W. Maass
International Conference on Information Systems (ICIS), 2024.

[9] **Plan Generation from Unstructured Documents through Transformer-based Extraction of Knowledge Graphs**
W. Maass, **C. K. Agnes**, A. Harig
European Conference on Information Systems (ECIS), 2024.

[10] **From Stateless to Adaptive: Dynamic Personalization for Conversational Language Models**
C. K. Agnes, W. Maass
NeurIPS, Women in Machine Learning Workshop, 2024.

[11] **Spatial-Economic Analysis for Optimal Electric Vehicle Charging Station Placement**
A. Maurya, **C. K. Agnes**, B. K. Baloch, S. S. Anwari, U. Butt
IEEE International Conference on Social and Sustainable Innovations in Technology and Engineering, 2024.

[12] **Improving Cardiovascular Health through AI-based Analysis of Genetic Risk Factors**
W. Maaß, O. Stegle, S. Janzen, A. M. Hickmann, **C. K. Agnes**, M. R. Rahman
Conference of the German AI Service Centers (KonKIS), 2024.

[13] **SNOOP Method: Faithfulness of Text Summarizations for Single Nucleotide Polymorphisms**
W. Maass, **C. K. Agnes**, M. R. Rahman, J. S. Almeida
AAAI Summer Symposium, 2023.

[14] **ExoSGAN and ExoACGAN: Exoplanet Detection Using Adversarial Training Algorithms**
C. K. Agnes, A. Naveed V., A. M. M. O. Chacko
arXiv preprint arXiv:2207.09665, 2022.

Technical Skills

Programming	Python, SQL, C++
ML & NLP	PyTorch, Hugging Face Transformers, scikit-learn
Data & Infrastructure	Pandas, NumPy, Docker, Git, SLURM, Weights & Biases
Methods	Representation learning, LLM-as-a-Judge, Knowledge graphs, Model evaluation

Languages

English • German (B1, Ongoing)